QUOIN: A Quantized Universal Observational Inference Network for Real-Time Transient Classification and Redshift Estimation

Anonymous submission

Abstract

The massive influx of time-domain data from the Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST) presents a critical need for highly precise automated transient characterization. Maximizing the scientific yield of these observations requires dual precision: the accurate taxonomic classification of supernova subtypes and the estimation of their corresponding redshifts. To address this, we introduce QUOIN (Quantized Universal Observational Inference Network), a novel three-tier encoder architecture engineered to optimize feature extraction from complex astronomical data streams. Rather than relying on uniform processing, QUOIN sequences data through three specialized layers to maximize prediction accuracy. First, a Selective State Space Model (SSM) layer filters the data stream, capturing long-range contextual dependencies and establishing a robust baseline representation of the astronomical signal. Second, an adaptive Dynamic Patching mechanism isolates and compresses local signal variations, precisely preserving critical, fine-grained localized features, such as narrow spectral lines and subtle phase transitions, that are vital for distinguishing subtle physical differences. Finally, these optimized feature patches are processed by a BERT-style Bidirectional Transformer Encoder to resolve complex, global contextual interactions across the entire sequence, yielding supernova classification and continuous redshift estimation. Evaluating our framework on simulated and real Rubin-era data streams, QUOIN demonstrates state-of-the-art accuracy in identifying fine-grained supernova subtypes and estimating redshifts, and particularly excels under low signal-to-noise conditions where traditional architectures struggle to resolve faint target features. QUOIN can be translated to other fields dealing with messy dependent data.

1 Introduction

Supernovae –the explosive deaths of stars– probe stellar evolution, nucleosynthesis, and cosmology. Type Ia supernovae serve as standardizable candles that revealed the universe's accelerating expansion (Riess et al. 1998; Perlmutter et al. 1999), winning the 2011 Nobel Prize in Physics. Core-collapse supernovae (Types Ib, Ic, II) provide insights into massive star evolution, the physics of neutron star and black hole formation, and the chemical enrichment of galaxies. Superluminous supernovae (SLSNe) challenge theoretical models with extreme luminosities ($> 10^{44}$ erg/s) likely powered by magnetar spin-down or circumstellar interaction.

Spectroscopic observations classify supernovae and measure their redshifts, revealing ejecta composition, velocity, temperature, and ionization via absorption and emission features. For instance, a strong Si II absorption feature near 6150 Å defines Type Ia supernova, whereas prominent hydrogen Balmer lines identify Type II supernovae (Filippenko 1997). Feature displacement from rest-frame wavelengths determines the redshift, providing the recession velocity and cosmological distance.

The unprecedented data volume of modern time-domain astronomy necessitates robust, automated spectral analysis workflows. The Vera C. Rubin Observatory's 10-year Legacy Survey of Space and Time (LSST; Ivezić et al. 2019) is projected to discover millions of transient events, creating a severe bottleneck for traditional spectroscopic follow-up. The massive scale of these astronomical datasets presents a critical machine learning challenge, underscoring the urgent need for accurate, real-time classification and redshift estimation pipelines capable of handling multidimensional 'dependent' astronomical data. Several interrelated challenges complicate the automated analysis of supernova spectra.

Heterogeneous data quality. Spectroscopic databases such as the Weizmann Interactive Supernova Data Repository (WISeREP; Yaron and Gal-Yam 2012) aggregate observations from dozens of observatories, instruments, and data reduction pipelines. The resulting spectra exhibit substantial variation in signal-to-noise ratio, wavelength coverage, spectral resolution, and flux calibration quality. Instrumental artifacts, atmospheric absorption residuals, and cosmic ray contamination further degrade data quality, particularly at the ultraviolet and near-infrared edges of spectral coverage.

Class imbalance. The observed population of spectroscopically classified supernovae is heavily dominated by Type Ia events, which are both intrinsically common and preferentially targeted for spectroscopic follow-up due to their cosmological importance. In the WISeREP dataset used in this work (Yaron and Gal-Yam 2012), Type Ia supernovae account for 57.9% of all spectra, while stripped-envelope supernovae (Type Ib/c) contribute only 11.7% and SLSNe contribute 3% (see Data section 3).

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Domain gap. Although synthetic spectral libraries such as SASSAFRAS (Wasserman 2026) generate hundreds of thousands of simulated supernova spectra with known ground truth, some of the statistical properties can inherently diverge from real data. This domain gap necessitates domain adaptation strategies. Implementing a sim-real co-training framework allows a network to simultaneously learn from both, extensive simulations and scarce real observations, by mapping them into a shared latent space to enforce domain-invariant representations and bridge the model-to-data divide (Maddukuri et al. 2025; Cheng et al. 2026).

Multi-objective prediction. Classification and redshift estimation are often treated as separate problems, yet they are intimately connected. The apparent wavelengths of spectral features depend on redshift, so knowledge of redshift aids classification, while the spectral type constrains the expected feature locations used for redshift measurement. Joint prediction frameworks that leverage this correlation can potentially outperform independent single-task models.

2 Related Work

Spectroscopic Classification of Supernovae Supernova typing has long been done by template cross-correlation. SNID (Blondin and Tonry 2007) correlates a flattened, log-rebinned query against a library of labeled templates and reports the type of the best matches; SNID–SAGE (Stoppa and Smartt 2026) retains that correlation engine and adds clustering over the ranked matches together with consolidated redshift and confidence estimates. Both return a redshift, but only as a by-product of the search: the correlation is evaluated over a grid of trial redshifts and whichever template wins supplies the answer, so the estimate degrades wherever the type is already uncertain. Both also inherit the type attractor effect, where an imbalanced library pulls low signal-to-noise queries toward the best-represented class (Blondin and Tonry 2007), and their cost grows with library size rather than staying fixed at inference.

DASH (Muthukrishna, Parkinson, and Tucker 2019) removed the library from the inference path, training a deep convolutional neural network (CNN) over rebinned spectra to predict type, age, redshift and host jointly. It is trained end to end for one fixed label set, and its redshift is an auxiliary head on a network optimized for classification. DASH lacks SLSN type of supernova in its native classification scheme. We therefore retrain it on our five-way taxonomy, so that SLSN-I is available to every method we compare.

Representation Learning for Astronomical Spectra Self-supervised models are displacing purpose-built networks elsewhere in astronomy. SpectraFM (Koblischke and Bovy 2024) pretrains on synthetic stellar spectra and transfers to unseen wavelength ranges with roughly one hundred labeled stars; AstroCLIP (Parker et al. 2024) and AstroM^3 (Rizhko and Bloom 2025) align spectra with images, photometry and metadata contrastively.

Two models target supernovae. Maven (Zhang et al. 2024) pretrains on 0.5M simulated events by aligning photometry with spectroscopy, fine-tunes on 4,702 ZTF supernovae, and reports both classification and redshift. Its signal is cross-modal agreement, so it requires paired photometry, and the authors report it does not decisively beat task-specific supervised models. DAEP (a diffusion autoencoder with perceivers) (Shen and Gagliano 2025) instead optimizes reconstruction with a Perceiver encoder and diffusion decoder, and assesses representations by linear probing on frozen features, including a three-way Ia / Ib-c / other split on ZTF spectra.

3 Data

WISeREP The Weizmann Interactive Supernova Data Repository (WISeREP, Yaron and Gal-Yam 2012) is a centralized SQL-based archive for supernova spectra and metadata from legacy and modern surveys. Using the WISeREP API, we bulk-downloaded all public spectra and key metadata (coordinates, redshift, and spectral type). We then refined our dataset by removing non-supernova objects and low-resolution spectra from the Spectral Energy Distribution Machine (SEDM). The final set contains $34{,}234$ spectra of $11{,}441$ unique supernovae, taken with 132 distinct instruments. We retain five parent types and merge subtypes into them: Ib/c aggregates Ib, Ic, Ic-BL, Ibn and related peculiars; Ia aggregates 91T-like, 91bg-like, Iax and other peculiar Ia; and II absorbs IIP and IIb.

The corpus is severely imbalanced across these types. SN Ia supplies $19{,}813$ spectra (57.9%) from $7{,}225$ objects, SN II $7{,}642$ (22.3%) from $2{,}572$, stripped-envelope SN Ib/c $3{,}997$ (11.7%) from $1{,}037$, SN IIn $1{,}740$ (5.1%) from 400, and SLSN-I, the rarest class, only $1{,}042$ (3.0%) from 211.

Synthetic Data We use publicly available synthetic spectra, SASSAFRAS (Wasserman 2026), of varying types of transient events (Ia, Ib, Ic, IcBL, IIb, IIn, KN, TDE, ILOT, 91bg, Iax, SLSN-I, CART), simulated using the Supernova Analysis Package (SNANA; Kessler et al. 2009). We aggregate these into the same five parent types as WISeREP and discard the four non-supernova transients (KN, TDE, ILOT, CART). SASSAFRAS data are simulated in uniform sample sizes for each transient type – rather than in quantities proportional to biased decisions for LSST spectroscopic follow-up – to mitigate the introduction of bias from an imbalanced training set. Instrument noise contributions are estimated using the exposure time calculators for the Gemini Multi-Object Spectrographs (GMOS) on the Gemini telescope, the Goodman High Throughput Spectrograph (GTHS) on the SOAR telescope, and the Low-Resolution Spectrograph on 4MOST. Spectra in SASSAFRAS are simulated with varying redshifts between 0.023 and 1, and phases ranging between -10 and $+50$ days relative to peak in the rest frame. Outputs include wavelength, flux, and flux error, MJD, phase, and redshift.

4 Method

We introduce Quoin (Quantized Universal Observational Inference Network), a novel three-tier encoder architecture engineered to optimize feature extraction from complex astronomical spectroscopy data streams. Quoin is a single backbone, pretrained once without labels on heterogeneous simulated and real spectra, that maps a spectrum to a compact summary representation and transfers to downstream tasks

by attaching a small head. The backbone realizes the three-tier design of Figure 1: a selective state-space model reads the full signal, a dynamic patching stage keeps only its most informative positions, and a bidirectional transformer contextualizes them. We evaluate this claim on three downstream tasks that reuse the same pretrained backbone: *observed-frame spectral classification* of supernovae into five types (SN Ia, SN Ib/c, SN II, SN IIn, SLSN-I), performed directly on the spectrum as observed and without any redshift information; *redshift prediction*, a continuous regression problem; and *rest-frame classification*, which composes the two by using the predicted redshift to remove the cosmological stretch before classifying.

Preprocessing

The spectra in our corpus come from instruments that differ in resolution, wavelength coverage, and flux calibration. As a preprocessing step, the raw spectra were denoised using a multiresolution analysis via the discrete wavelet transform (Mallat 1989), effectively isolating high-frequency noise from the underlying spectral signals, while preserving intrinsic spectral structures. This step is followed by standardizing each resulting spectrum to have a zero flux mean and unit variance computed as

$$\tilde{f}_i \;=\; \frac{f_i - \mu_f}{\sigma_f}, \tag{1}$$

where μ_f and σ_f denote the mean and standard deviation of the individual spectrum's flux. We then resample all spectra onto a common wavelength support and fixed length N, with a boolean mask marking the valid positions.

The Quoin Encoder

The Quoin encoder embeds a spectrum as a token sequence, reads it in linear time with a selective state-space model (SSM), retains only its most informative positions, and contextualizes those positions with a bidirectional transformer (Figure 1).

Flux tokenization. Motivated by masked language modeling (Devlin et al. 2019), we treat flux as a discrete token rather than a scalar. Each standardized value is assigned to one of V uniform bins spanning a fixed range $[f_{\min}, f_{\max}]$,

$$b(f) \;=\; \mathrm{clip}\!\left(\left\lfloor \frac{f - f_{\min}}{f_{\max} - f_{\min}}\,(V-1)\right\rfloor,\, 0,\, V-1\right), \tag{2}$$

and mapped through a learned embedding table $E \in \mathbb{R}^{V\times d}$ to a token embedding. Extreme values saturate harmlessly into the tail bins, and self-supervised pretraining reduces to a tractable classification problem over a finite vocabulary (Eq. 6).

Physical-wavelength positions. Position is encoded from the *physical* wavelength rather than the array index. A sinusoidal feature of λ_i is passed through a small MLP and added to the flux token,

$$\mathbf{x}_i^{(0)} \;=\; E_{b(\tilde{f}_i)} + \mathrm{MLP}\big(\,[\sin(\omega_k\lambda_i), \cos(\omega_k\lambda_i)\,]_k\,\big). \tag{3}$$

Because position is a function of wavelength alone, a line at a given λ maps to the same representation regardless of the instrument's sampling.

Selective state-space reader. The full-length token sequence is read by a stack of selective state-space (Mamba) blocks (Gu and Dao 2023). A hidden state $\mathbf{h}_i$ is propagated along the wavelength axis with an input-dependent step Δ_i,

$$\mathbf{h}_i = \bar{\mathbf{A}}_i\,\mathbf{h}_{i-1} + \bar{\mathbf{B}}_i\,x_i, \quad y_i = \mathbf{C}_i\,\mathbf{h}_i, \quad \bar{\mathbf{A}}_i = \exp(\Delta_i\mathbf{A}), \tag{4}$$

where x_i is the token embedding of Eq. 3 and y_i the output the block emits at position i, the quantity scored in Eq. 5. The underlying system is continuous in wavelength, with a learned diagonal transition $\mathbf{A}$ that is independent of the input; the operators $\bar{\mathbf{A}}_i$ and $\bar{\mathbf{B}}_i$ are its transition and input map discretized at the step Δ_i, so the step controls how strongly the current token enters the state and how much of the previous state survives. The scan is $O(N)$ in sequence length and gives exactly the inductive bias a spectrum calls for: sharp line features are written strongly into the state, while flat continuum passes through nearly unchanged.

Dynamic salience patching. Most positions carry continuum and little information, so we prune the sequence before the quadratic-cost transformer. A parameter-free patcher scores each position by the squared deviation of its SSM embedding $\mathbf{y}_i$ from a local moving mean over a window W,

$$s_i \;=\; \Big\|\,\mathbf{y}_i - \tfrac{1}{W}\sum_{|j-i|\le W/2}\mathbf{y}_j\,\Big\|_2^2, \tag{5}$$

masks padded positions, keeps the K highest-scoring positions, and restores them to wavelength order.

Bidirectional transformer and summary token. The K retained tokens, prepended with a learnable summary token `[CLS]`, pass through a pre-norm bidirectional transformer encoder (Devlin et al. 2019). The output at the summary position, $\mathbf{h} \in \mathbb{R}^d$, is the representation consumed by every task head.

Self-supervised pretraining. The backbone is pretrained by masked modeling: a fraction of input tokens are replaced by a mask symbol, and a lightweight head predicts their original flux bins $b(\tilde{f}_i)$ from context under a cross-entropy loss over the masked set $\mathcal{M}$,

$$\mathcal{L}_{\mathrm{MLM}} = -\sum_{i\in\mathcal{M}} \log p_\theta\big(b(\tilde{f}_i)\,\big|\,\mathrm{context}\big). \tag{6}$$

Pretraining co-trains on simulated and real spectra, and the resulting weights initialize every downstream task.

Task 1: Observed-Frame Spectral Classification

The first task classifies a spectrum exactly as it is observed. No redshift is supplied and none is inferred: the input is the observed-frame spectrum, and a small head maps the summary representation $\mathbf{h}$ to logits over the five supernova types. The classifier must therefore recognize a type across the whole family of redshifted appearances that type can take – the burden that Section 4 later removes by de-redshifting first. This makes Task 1 the setting that requires the least auxiliary information at inference, and the natural point of comparison for methods that classify without a redshift estimate.

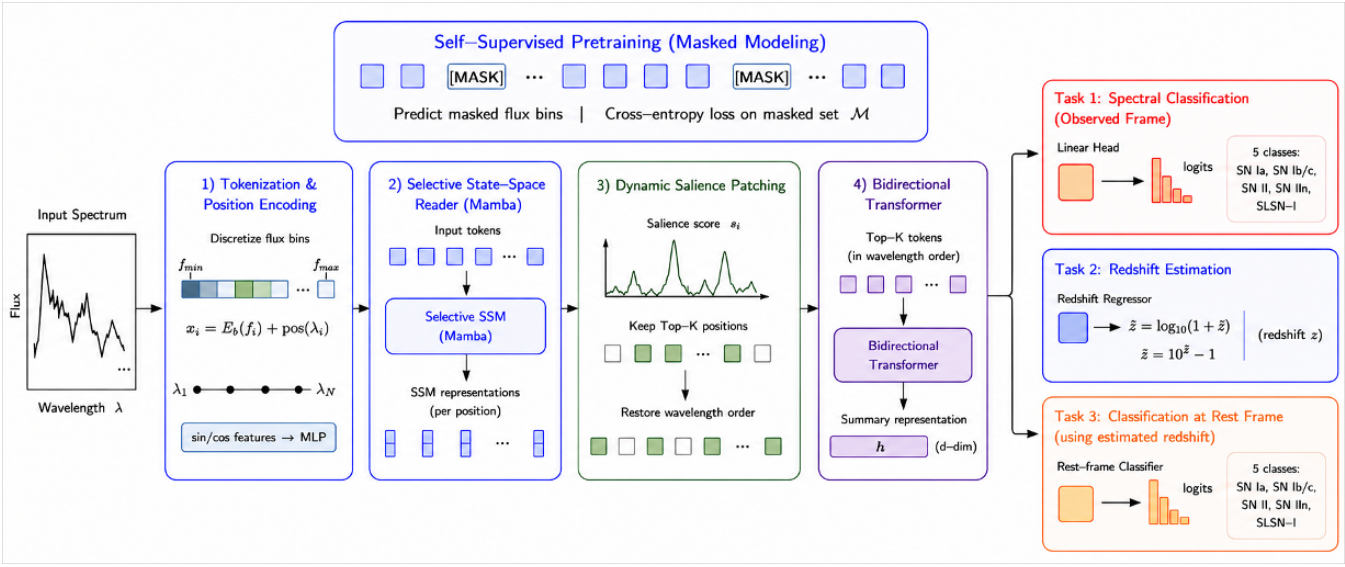


Figure 1: Overview of the QUOIN pipeline.

The central difficulty is a data asymmetry: real labeled spectra are scarce and severely class imbalanced, while simulations are abundant and cover the rare classes but are domain-shifted. Domain co-training (Maddukuri et al. 2025; Cheng et al. 2026) reconciles the two. Each mini-batch mixes simulated and real spectra, and we minimize a convex combination of the two cross-entropies,

$$\mathcal{L}_{\mathrm{cls}} = \alpha\, \mathrm{CE}_{\mathrm{sim}} + (1-\alpha)\, \mathrm{CE}_{\mathrm{real}}, \tag{7}$$

together with class weights that up-weight the rare types. The simulated stream supplies rare-class diversity, the real stream anchors the decision boundary to the observed domain, and the mixing weight α makes the trade-off explicit; we characterize its effect in Experiments section 5.

Task 2: Redshift Prediction

The second task changes the output type: from a categorical label to a continuous physical quantity, using the same backbone and the same pretrained weights. Cosmic expansion multiplies every observed wavelength by $(1+z)$, so recovering z amounts to measuring how far a spectrum's features have been displaced from their laboratory positions. A regression head maps the summary representation $\mathbf{h}$ to a scalar prediction $\hat{y}$, trained against the log-space target $y = \log_{10}(1+z)$ under a squared-error loss. For a mini-batch $\mathcal{B}$,

$$\mathcal{L}_z(\mathcal{B}) = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \big(\hat{y}_i - \log_{10}(1+z_i)\big)^2, \qquad \hat{z}_i = 10^{\hat{y}_i} - 1, \tag{8}$$

where the log parameterization stabilizes training over the wide dynamic range of z and keeps $\hat{z}_i > -1$ by construction. The regressor is co-trained as in Eq. 7, weighted toward the real spectra whose low-redshift distribution matches the test domain; the simulated stream, which spans a much broader redshift range, supplies the coverage needed to learn the mapping at all.

This task also motivates the physical-wavelength positional encoding of Eq. 3. Because position is a function of λ rather than array index, a redshift acts on the representation as a coherent translation of the positional field, which is the structure the regressor must detect.

Task 3: Rest-Frame Classification

The third task composes the first two: the prediction of Task 2 becomes the input transformation for Task 1. Because a class appears not as one template but as a family of redshifted copies, a classifier reading observed-frame spectra must implicitly marginalize over z. Supplying the redshift instead removes that burden. Given a predicted redshift, the spectrum is moved to its rest frame by rescaling wavelengths,

$$\lambda_{\mathrm{rest}} = \frac{\lambda_{\mathrm{obs}}}{1+\hat{z}}, \tag{9}$$

which aligns each class's diagnostic features to common wavelengths before the classification head of Section 4 is applied.

Crucially, the rest-frame classifier is trained on spectra deredshifted with the *predicted* redshift, not the catalog value. Training and test inputs therefore share a distribution, and the classifier learns to absorb the regressor's systematic errors rather than encountering them for the first time at inference. In Results, we evaluate this fully predictive pipeline against the observed-frame classifiers of Task 1.

5 Experiments

Experimental Setup

Baselines. No baseline is used off the shelf: each is either rebuilt on our data or given a reference library drawn from our training split, so differences reflect the model rather than the data it was fitted to. Because released weights and template libraries are fitted to catalogues that overlap our test objects, using them as published would leak test labels

into every baseline. We therefore retrain each one, and name them accordingly: *-style* denotes the published architecture trained from scratch on our split, so the design is the original authors' and the parameters are ours. **SNID (rebuilt templates)** (Blondin and Tonry 2007; Stoppa and Smartt 2026) is the one baseline whose released code we keep unchanged: the cross-correlation engine is the distributed one, and only its reference library is replaced. We build 999 templates ($\approx$ 200 per class) from our training split alone, deredshifted, continuum-flattened, and rebinned onto SNID's log-wavelength grid so templates and queries are prepared identically. **DASH-style CNN** reproduces the published DASH architecture — three Conv1d/BatchNorm/MaxPool blocks into two fully connected layers, 1.4M parameters — trained from scratch on our split over the same preprocessed spectra as QUOIN, resampled onto a uniform 1024-point grid. It receives no redshift. **DAEP-style encoder** pairs the authors' released Perceiver encoder with our own classification and redshift heads, trained from scratch on our data.

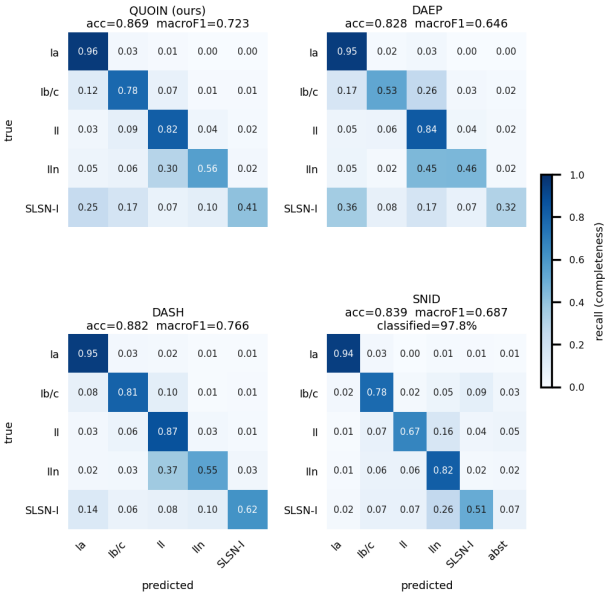


Figure 2: Observed-frame classification under WISeREP-only training, as *completeness*: cells are row-normalized, so each row gives the recall of a true class and sums to one. "abst" is the abstention column for SNID, which declines to classify 2.2% of spectra.

Performance Metrics. *Classification.* The five types are severely imbalanced in WISeREP, so accuracy is dominated by SN Ia and is not on its own informative. We report it together with the macro-averaged F_1, which weights every type equally. At the class level we give both normalizations of the confusion matrix, since neither suffices alone: *completeness* (recall) and *purity* (precision) for class c,

$$\mathrm{Comp}_c = \frac{n_{cc}}{\sum_k n_{ck}}, \qquad \mathrm{Pur}_c = \frac{n_{cc}}{\sum_k n_{kc}}, \tag{10}$$

where n_{ck} counts spectra of true class c predicted as k. A method can inflate completeness on a rare class simply by

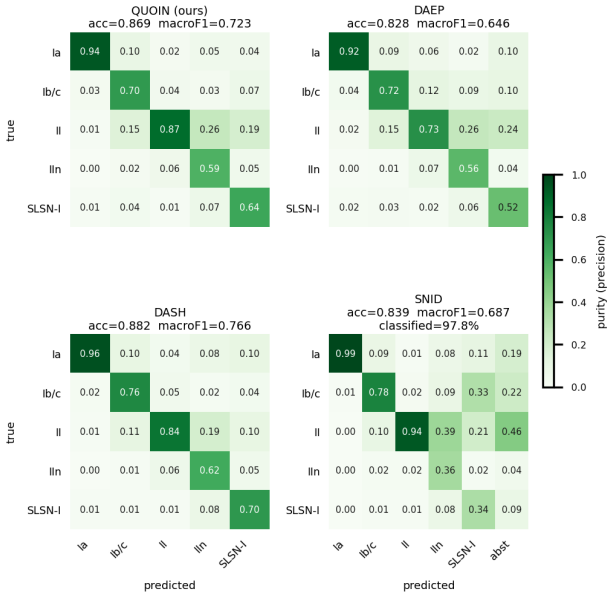


Figure 3: The same predictions as Figure 2, as *purity*: cells are column-normalized, so each column gives the precision of a predicted class and sums to one.

over-assigning its label; only purity exposes this, and the macro-F_1, averaging the per-class harmonic mean, penalizes it. Where a template method declines to classify a spectrum we report *coverage*, the fraction assigned a label, and count each abstention as an error rather than dropping it from the denominator, so that coverage and accuracy cannot be traded against one another.

Redshift. The regressor is trained in $\log_{10}(1+z)$ but evaluated in z after inverting Eq. (8), so the numbers are directly comparable to catalog redshifts. Because the estimators differ less in the size of their errors than in whether those errors have a preferred direction, we pair the mean absolute error with three measures built on the $(1+z)$-normalized residual $\delta_i = (\hat{z}_i - z_i)/(1+z_i)$, where $\delta > 0$ denotes over-prediction:

$$\begin{aligned} \mathrm{MAE} &= \frac{1}{n}\sum_{i=1}^{n} |\hat{z}_i - z_i|, \\ \mathrm{bias} &= \mathrm{med}(\delta), \\ \sigma_{\mathrm{NMAD}} &= 1.4826\,\mathrm{med}(|\delta - \mathrm{med}(\delta)|)\,, \\ f_+ &= \frac{1}{n}\,|\{\, i : \hat{z}_i > z_i \,\}|\,. \end{aligned} \tag{11}$$

MAE summarizes typical accuracy, $\mathrm{med}(\delta)$ and f_+ capture systematic offset in magnitude and in direction, and σ_{NMAD} isolates the scatter of the well-behaved bulk from the tail. An unbiased estimator gives $f_+ = 1/2$, and reporting it alongside $\mathrm{med}(\delta)$ guards against an offset hidden by over- and under-predictions cancelling. We normalize by $(1+z)$ rather than by z because the sample reaches $z \sim 10^{-3}$, where dividing by z turns an ordinary absolute error into a four-digit percentage.

Model	MAE	$\mathrm{med}(\delta)$	σ_{NMAD}	f_+
Quoin	**0.0172**	**+0.0020**	**0.0114**	**56.3%**
DAEP-style	0.0216	+0.0072	0.0142	70.1%
DASH-style	0.0425	+0.0287	0.0269	92.8%

Table 1: Redshift estimation, evaluated in z. All three estimators over-predict ($\mathrm{med}(\delta) > 0$, $f_+ > 1/2$), but Quoin does so by an order of magnitude less than DASH-style and is closest to the unbiased $f_+ = 1/2$. Lower is better for MAE, $|\mathrm{med}(\delta)|$ and σ_{NMAD}; $f_+ = 50\%$ is ideal.

Training. Quoin is a BERT-base encoder (d=768, 12 heads, 12 layers, FFN 3072, dropout 0.2) augmented with four state-space layers and dynamic patching that retains 256 tokens from sequences of length 1024. All models were trained on a single NVIDIA A100-SXM4 80 GB GPU with AdamW (β_1=0.9, β_2=0.999, ϵ=10^{-8}, weight decay 10^{-2}), a peak learning rate of 10^{-4} under 1000 linear warmup steps followed by cosine annealing, and gradient clipping at 0.5. Co-training draws each step from a mixed sampler combining a simulated mini-batch of 128 with a real mini-batch of 32; WISeREP-only fine-tuning uses a single batch of 128. Class imbalance is handled with inverse-square-root class weights and label smoothing.

Observed-Frame Spectral Classification

We first classify observed-frame spectra directly, with no redshift information, training on WISeREP alone. Because a class can be recovered simply by over-assigning its label, no single normalization of the confusion matrix ranks the methods on its own, so we report both: completeness in Figure 2 and purity in Figure 3. SNID shows why this is necessary. It reaches the highest SN IIn completeness of any approach (0.822) on only 0.363 purity, so two thirds of its SN IIn calls are wrong, and its SN II behavior is the mirror image, 0.937 purity against 0.669 completeness.

Under WISeREP-only training (Figures 2 and 3) accuracy and macro-F_1 are close across the learned methods: 0.882 and 0.766 for the DASH-style CNN, 0.869 and 0.723 for Quoin, 0.839 and 0.687 for SNID, and 0.828 and 0.646 for the DAEP-style encoder. This setting is the one most favorable to a compact task-specific classifier, since it fixes a single domain, a few thousand labeled spectra and a five-way taxonomy, and it leaves a pretrained backbone with little extra to exploit.

Co-training changes the picture. Adding the simulated stream supplies the rare-class diversity WISeREP lacks, and it is the setting a pretrained backbone is built for, since the two domains share a representation rather than a label set. Figure 4(a) reports completeness and Figure 5(a) purity for the three learned models; SNID is omitted because its template library is fixed at construction and admits no co-training stage. Quoin leads on both aggregate measures, at 0.859 accuracy and 0.726 macro-F_1 against 0.852 and 0.694 for the DASH-style CNN and 0.839 and 0.629 for the DAEP-style encoder.

On the two rarest classes the completeness of Quoin and the DASH-style CNN is nearly identical, 0.587 against 0.583 on SN IIn, and on SLSN-I the CNN is in fact ahead, 0.534 against 0.494. Purity separates them: 0.620 against 0.491 on SN IIn, and 0.593 against 0.372 on SLSN-I. The CNN therefore recovers rare classes in part by over-assigning their labels, so nearly two thirds of its SLSN-I calls are wrong, while Quoin recovers the same objects at a precision that survives the harmonic mean. Accuracy is blind to this, since both rare classes together are a small fraction of the test set; macro-F_1 is not, which is why the ordering the two measures induce differs in magnitude even where it agrees in sign.

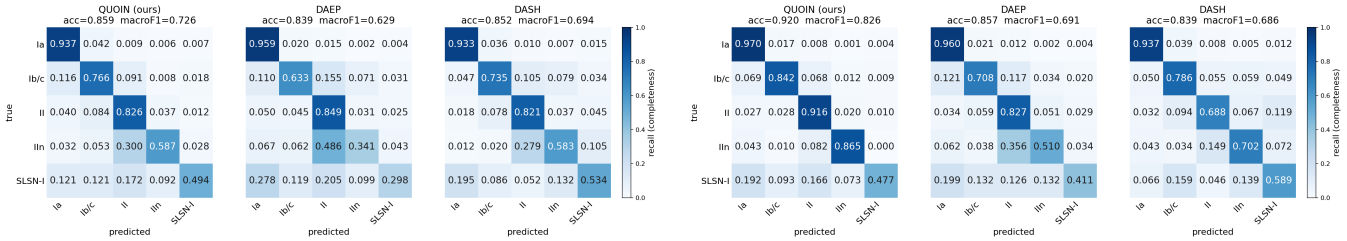
The dominant failure mode is shared by every method and is astrophysically sensible rather than arbitrary: SN IIn are absorbed into SN II, and SLSN-I are absorbed into SN Ia and SN Ib/c. These are exactly the pairs whose observed-frame continua are most similar once redshift is unknown, since without it a classifier cannot separate an intrinsic difference in spectral shape from a displacement of the same shape along the wavelength axis. This is the burden that the rest-frame treatment of Section 5 removes.

Redshift Prediction

Figure 6 plots predicted against true $\log_{10}(1+z)$ for Quoin and the two learned baselines, and Table 1 reports the four summary statistics of the predictions. SNID is not included: its redshift is read off the template correlation grid rather than produced by a regression head, and it abstains on part of the sample, so it is not commensurable with the three regressors. All three over-predict, but by very different margins. Quoin places 56.3% of the sample above the true value, close to the 50% expected of an unbiased estimator, against 70.1% for DAEP-style and 92.8% for DASH-style, which over-predicts nearly every object it sees. Table 1 confirms this with the median normalized residual, which gives the same ordering (+0.0020, +0.0072, +0.0287), so the conclusion does not depend on which definition of bias is adopted. Figure 6 shows the same fact geometrically: DASH-style sits almost entirely above the identity line, DAEP-style leans above it, and Quoin scatters about it.

The size of the error follows the same order. In Table 1, Quoin reaches an MAE of 0.0172 against 0.0216 for DAEP-style and 0.0425 for DASH-style, and the ranking is unchanged under σ_{NMAD} (0.0114, 0.0142, 0.0269), so the advantage is not an artifact of a few catastrophic failures; the bulk of the distribution is tighter as well. Because $z \lesssim 0.1$ across this sample, $1+z \approx 1$ and $\mathrm{med}(\delta)$ can be read directly against the MAE. Quoin's bias is then roughly one eighth of its MAE, so most of its error is symmetric scatter that cancels in the mean, whereas DASH-style's is two thirds of its MAE, so its error is predominantly a systematic offset rather than noise. The distinction matters downstream: scatter averages away with sample size, while an offset does not, and it propagates into any distance or luminosity derived from the estimate.

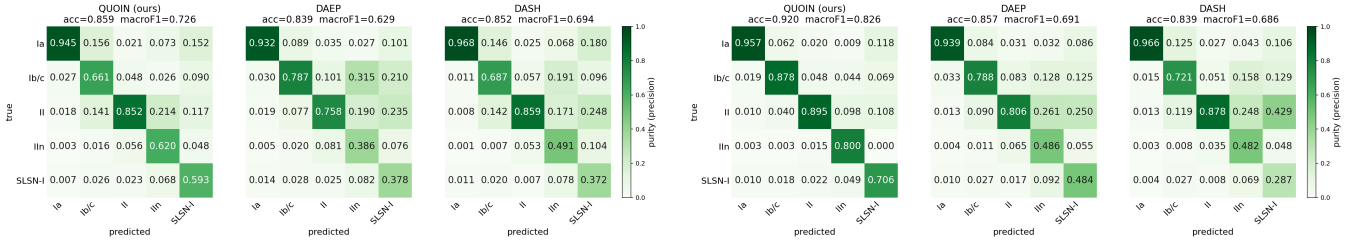
What drives the ordering is the input representation rather than model capacity. DASH-style consumes flux resampled onto a fixed grid, so the wavelength axis survives only implicitly as bin position, whereas Quoin and DAEP-style both receive wavelength and flux as an explicit pair. Redshift is determined by *where* features fall in wavelength, so an architecture that discards that axis is handicapped for the task, and



(a) SimReal co-training, observed frame

(b) Rest frame, predicted redshift

Figure 4: Confusion matrices, row-normalized so each row gives the completeness of a true class. (a) Observed-frame classification under SimReal co-training. (b) The same classes after de-redshifting with the predicted redshift.



(a) SimReal co-training, observed frame

(b) Rest frame, predicted redshift

Figure 5: *Purity* for the same predictions as Figure 4 (column-normalized, each column sums to one).

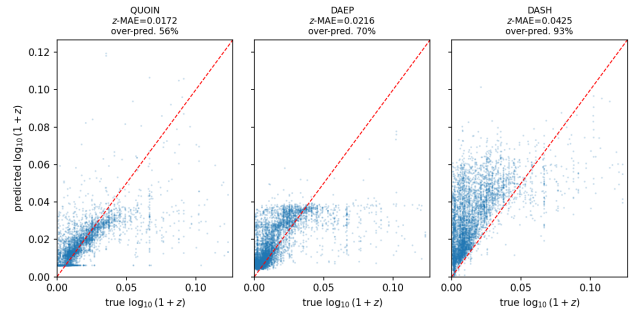


Figure 6: Predicted versus true redshift on the held-out set. The dashed line is the identity.

its MAE is $2.5\times$ worse than ours. The residual gap between Quoin and DAEP-style, which share the explicit encoding, is smaller. We attribute it to the physical-wavelength positional encoding of Eq. 3: because position is a function of λ rather than array index, a redshift acts on the representation as a coherent translation of the positional field, which is exactly the structure the regressor must detect.

Rest-Frame Classification with Predicted Redshift

The two preceding components are now combined. The redshift predictor estimates $\log_{10}(1+z)$ from the observed spectrum, the wavelength axis is de-redshifted using that *predicted* value, and the rest-frame classifier assigns a type to the resulting spectrum. No ground-truth redshift enters this pipeline at any point; true redshifts are used only for the diagnostic in Section 5. The rest-frame classifier is trained on spectra de-redshifted by the same predicted quantity, so training and test conditions match.

Figure 4(b) shows the effect on the confusion structure, directly beside the observed-frame co-training panel in (a). Quoin reaches 0.920 accuracy and 0.826 macro-F_1 in the rest frame, ahead of both baselines on either measure (0.857 and 0.691 for the DAEP-style encoder, 0.839 and 0.686 for the DASH-style CNN). The SN IIn/SN II confusion that dominates the observed-frame matrices is substantially resolved once spectra are placed in a common rest frame, which is the mechanism we claim: the classifier no longer has to disentangle intrinsic spectral shape from redshift-induced displacement.

6 Conclusion

In this work, we developed a new 3-tier encoder model designed to directly model the dependent structure between wavelength and flux pairs for supernova spectral analysis. Our classification and redshift estimation results demonstrate that the architecture of Quoin consistently outperforms all established baselines—including SNID, DASH-style and DAEP-style encoders – by delivering significantly higher redshift precision and classification accuracy. While Quoin achieves robust performance across standard classifications, its application to SLSNe remains a work in progress. For these rare events, Quoin achieves exceptionally high classification purity (compared to baselines), but its completeness remains low. Future work will include focusing on expanding training libraries, optimizing the model to capture a broader range of unique SLSN spectral traits without losing precision. Beyond astrophysics, our Quoin architecture's ability to precisely align and interpret irregular, dependent 'messy' data holds broader promise for e.g., biomedical datasets, offering a robust framework for decoding complex, high-dimensional signals.

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